

Rearrangement of 3D genome organization in breast cancer epithelial - mesenchymal transition and metastasis organotropism

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eLife Assessment

This **valuable** study explores the role of spatial genome organization in oncogenic transformation, addressing an ambitious and significant topic. The authors have assembled comprehensive datasets from various subtypes of localized and lung-metastatic breast cancer cells, as well as from healthy and cancerous lung cells. They identified switching patterns in the 3D genome organization of lung-metastatic breast cancer cells, revealing a reconfiguration of genome architecture that resembles that of lung cells. This provides **solid** evidence with significant biomedical implications for epigenetic regulation in both normal physiology and disease.

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Abstract

Breast cancer cells exhibit organotropism during metastasis, showing preferential homing to certain organs such as bone, lung, liver, and brain. One potential explanation for this organotropic behavior is that cancer cells gain properties that enable thriving in certain microenvironments. Such specific metastatic traits may arise from gene regulation at the primary tumor site. Spatial genome organization plays a crucial role in oncogenic transformation and progression, but the extent to which chromosome architecture contributes to organ-specific metastatic traits is unclear. This work characterizes chromosome architecture changes associated with organotropic metastatic traits. By comparing a collection of genomic data from different subtypes of localized and lung metastatic breast cancer cells with both normal and cancerous lung cells, we find important trends of genomic reorganization. The most striking differences in 3D genome compartments segregate cell types according to their epithelial vs. mesenchymal status. This EMT compartment signature occurs at genomic regions distinct from transcription-defined EMT signatures, suggesting a separate layer of regulation. Specifically querying organotropism, we find 3D genome changes consistent with adaptations needed to survive in a new microenvironment, with lung metastatic breast cells exhibiting compartment switch

signatures that shift the genome architecture to a lung cell-like conformation and brain metastatic prostate cancer cells showing compartment shifts toward a brain-like state. TCGA patient data reveals gene expression changes concordant with these organ-permissive compartment changes. These results suggest that genome architecture provides an additional level of cell fate specification informing organotropism and enabling survival at the metastatic site.

Significance

Computational analysis of a cohort of cancer cell lines reveals some 3D genome spatial compartment changes are associated with transitions in cancer cell state that relate to metastasis (EMT) and others reflect the characteristics of the metastatic organ context.

Introduction

Breast cancer is the leading malignant disease in women worldwide. It is highly heterogeneous; based on molecular characteristics it can be divided into four broad subgroups: luminal A (estrogen receptor ER+ progesterone receptor PR+/- human epidermal growth factor receptor 2 HER2-), luminal B (ER+ PR+/- HER2+), HER2-enriched (ER- PR- HER2+), and triple-negative/basal (ER- PR- HER2-) (1). About twenty to thirty percent of early-stage breast cancer patients have a chance to ultimately develop distant metastasis, which accounts for ninety percent of all breast cancer-related deaths (2). Overall, metastasis implies the dissemination of cancer cells from the primary tumor site into the circulation and later extravasation to a new site which leads to the formation of a secondary tumor. However, the organs affected by breast cancer metastasis vary remarkably, depending on the disease subtype, molecular features, primary tumor location, and secondary site microenvironment. This phenomenon is known as ‘metastasis organotropism’ (3). For example, breast cancer cells can metastasize to the bone, lung, liver, and brain. However, bone metastasis accounts for a major percentage of metastatic cases compared to other organs (4). Moreover, luminal A and basal subtypes of breast cancer show a higher incidence of bone and lung metastasis, respectively, whereas luminal B preferentially metastasizes to both organs (5). Organotropic behavior is congruent with the ‘seed and soil’ hypothesis, according to which a cancer cell (seed) having specific intrinsic abilities will only be able to survive in a host local microenvironment (soil) that is favorable to it (6).

Secondary organ colonization by circulating tumor cells is a complex process, ruled by dynamic and evolving interactions between cancer cells and the host microenvironment. In most situations, the microenvironment is unfavorable, and the disseminated cells fail to survive in that ‘soil’. Despite the low survival rate of cancer cells during the metastatic colonization process, cells with the intrinsic capabilities to interact with the host microenvironment can survive and proliferate in the secondary site (7). Alternatively, cells might arrive at the secondary site and remain dormant under the influence of the local microenvironment (8–11) slowly acquiring characteristics that would later allow their proliferation. Importantly, the microenvironment surrounding primary breast tumors seems to condition the cells to specific metastatic profiles. For example, if the primary tumor stroma is enriched in transforming growth factor beta (TGFB), lung metastasis prevails through the induction of angiopoietin-like 4 (12). In contrast, a tumor stroma enriched with mesenchymal cytokines like C-X-C motif chemokine 12 (CXCL12) and insulin-like growth factor 1 (IGF1) predisposes primary cancer cell population to home into the bone marrow (13). How these different reactive microenvironments condition the origin of specific metastatic traits still poses an outstanding question.

The current view in the cancer biology field is that transcriptional gene signatures associated with organ-specific metastatic traits are regulated in part by cues from the reactive microenvironment that accompanies primary tumorigenesis (12–15). However, whether regulating those genes involves changes in 3D genome organization is still an open question. Recent advancements in imaging and molecular biology techniques have shown that the genome organizes in a non-random multi-layered fashion inside the nucleus, and each layer of organization contributes both independently and synergistically to gene expression regulation (16). At a large scale, the reorganization of genome compartments, in which euchromatic and heterochromatic genomic regions are spatially segregated, is also found to be associated with cancer progression (17–19). It has been shown that as breast cancer becomes endocrine-resistant, the change in compartment organization is accompanied by gene expression changes and differential estrogen receptor binding (17). Other studies have also found that an unintended consequence of various drug treatments and therapies is the rearrangement of chromatin architecture at different levels (17,20,21). Though these types of studies are essential for elucidating the association between the reorganization of the genome in three dimensions and oncogenic transformation, they do not consider the genome organization changes experienced by the metastatic cell that occur as a result of the interaction with the microenvironment of the secondary site. Genome spatial compartmentalization is strongly associated with cell type definition: genomic regions segregated into the A (euchromatic) or B (heterochromatic) compartments differ among cell types and tissue origin (22). These compartmentalization profiles help encode stable cell fate gene expression (23–25). Therefore, we hypothesize that similar to gene expression, metastatic cancer cells experience 3D genome structure rearrangements that mimic the compartment signatures of their new host microenvironment. To test this hypothesis, here we focused on the breast cancer-lung metastasis system. We compared genome organization data from different types of breast epithelial cell lines (normal, localized cancer, and metastatic) with normal and cancerous lung cells. Our analysis revealed that metastatic breast cancer cells have a higher proportion of lung-specific spatial compartment changes than localized breast cancer cells. We find similar brain microenvironment-specific compartment changes in prostate cancer cells that have the potential to metastasize to brain. These distal-site concordant compartment shifts often correspond to changes in gene expression in both cell lines and TCGA patient samples. However, we find that 3D genome changes do not always correlate with gene expression changes. Instead, particularly across the epithelial mesenchymal transition (EMT) spectrum, we find a set of genes that change expression without needing spatial reorganization and then a distinct set of epithelial and mesenchymal phenotype genes that change their spatial compartment without altered gene expression. We hypothesize that spatial genome organization facilitates immediate changes in gene expression in certain loci but can also poise certain regions for future activation in a way that favors EMT and adaptation to metastatic organ microenvironments.

Results

Modeling the breast cancer lung metastasis system

To investigate whether breast cancer spatial genome organization showed rearrangements specific to a distant organ metastatic microenvironment, we needed to assemble genome spatial compartment data from different types of normal and cancerous cells for both breast and a secondary organ. Since most of breast cancer subtypes show a preference for lung metastasis, we first focused on this metastatic site (26). We selected a cohort of breast and lung cell lines, both normal and cancerous, for which publicly available genome organization (Hi-C) are available (Supplementary Table 1). A total of 15 cell lines with available data, including primary cells and cancer cell lines from both organs, were selected. A classification of the cells used in this study based on their malignancy status and other molecular signatures is shown in **Fig. 1a**. Briefly, for the breast cancer system we considered both HMEC and MCF10A as normal (non-tumorigenic) breast epithelial cells and placed them at the starting node of the model. Cancer cell lines were

then divided into three subcategories based on breast cancer molecular signatures: luminal (ZR751, BT474, ZR7530, MCF7 and T47D), HER2-enriched (HCC1954 and SKBR3) and triple-negative (HCC70, BT549 and MDA-MB-231). We then assigned cell lines to “localized” or “metastatic” status based on the site from which they were originally collected in the patient (see details in Supplementary Table 1)([27](#), [28](#), [29](#), [30](#)). Specifically, within the luminal subtype, ZR751 and BT474 were designated as localized cancers and the remaining three cell lines, ZR7530, MCF7, and T47D were used as models for ascites, lung, and lung metastatic breast cancers respectively. Similarly, for the HER2-enriched subcategory, HCC1954 and SKBR3 were considered localized and lung metastatic cancers, respectively. Finally, in the case of the triple-negative subtype, HCC70 and BT549 were used to model localized cancer whereas MDA-MB-231 represents lung metastatic cancer. For the lung portion of the system, HTBE primary cells represent normal lung epithelium and both the A549 and H460 cell lines represent localized lung cancer. The valuable MetMap study in mouse has measured the organs that can be colonized by human-derived cancer cell lines when they are injected into mice.[\(31\)](#) The cell lines we consider here may have different or additional “potential” metastatic sites in this type of mouse model. But, it is known that species differences can lead to divergent organotropism in mouse models as compared to human patients.[\(32\)](#), [33](#) Therefore, we use the site of collection as compared to the site of origin in the human patient as our reference point for primary and secondary site organs.

Subtype and epithelial-mesenchymal state-specific signatures are present in breast cancer compartmental organization

We compared the supervised ordering of the cell lines described above with the ordering derived from unsupervised analysis of genome-wide A/B compartment organization data. Compartment identity strengths were calculated using eigenvector decomposition on the 250 kb binned Hi-C data for each cell line (Methods). In this analysis a positive value signifies a euchromatic A compartment region, while a negative value represents a heterochromatic B compartment association.

When we applied hierarchical clustering on the genome-wide compartment data, we observed a grouping of cells that was congruent with the cell ordering constructed based on molecular and clinical features (**Fig. 1b**). For example, all the normal cell types, including both the breast and lung systems, were clustered together. Similarly, cells from the luminal subtype and the lung carcinoma cells were grouped in their respective clusters. However, we also observed a few interesting exceptions from the clustering result. MDA-MB-231, a breast cancer cell line isolated from lung pleural effusion, was grouped with the lung carcinoma cell lines, suggesting its compartment organization shares more features with lung cancer than do other breast cancer cell lines. We also found that HCC70 and HCC1954, which were both derived from localized cancer, clustered relatively close to the normal epithelial cells. Collectively, this unsupervised analysis shows that the different subtypes of cancer exhibit distinctive 3D genome organization features at the compartment level (**Fig. 1b**), as previous studies suggested [\(18\)](#).

Performing principal component analysis (PCA) on the genome-wide compartment data revealed similar trends (**Fig. 1c**). Interestingly, the ordering of cell types along the first principal component (PC1) captured a transition between epithelial and mesenchymal-like cells. Cells characterized as highly epithelial were found on the left extreme of the PC1 axis, and as we move towards the right along PC1, the cells acquire a mesenchymal phenotype. For example, MDA-MB-231 cells are derived from healthy cells of epithelial origin but have undergone an epithelial-mesenchymal transition and show a mesenchymal-like phenotype [\(32\)](#), [33](#). In our Hi-C principal components analysis, this cell line has a highly positive PC1 value. In contrast, MCF7 cells maintain a more epithelial morphology and are found at the opposite end of the PC1 spectrum from MDA-MB-231. The organization of cell lines along PC1 from spatial compartmentalization matches quite well with the previously characterized epithelial-mesenchymal classification of

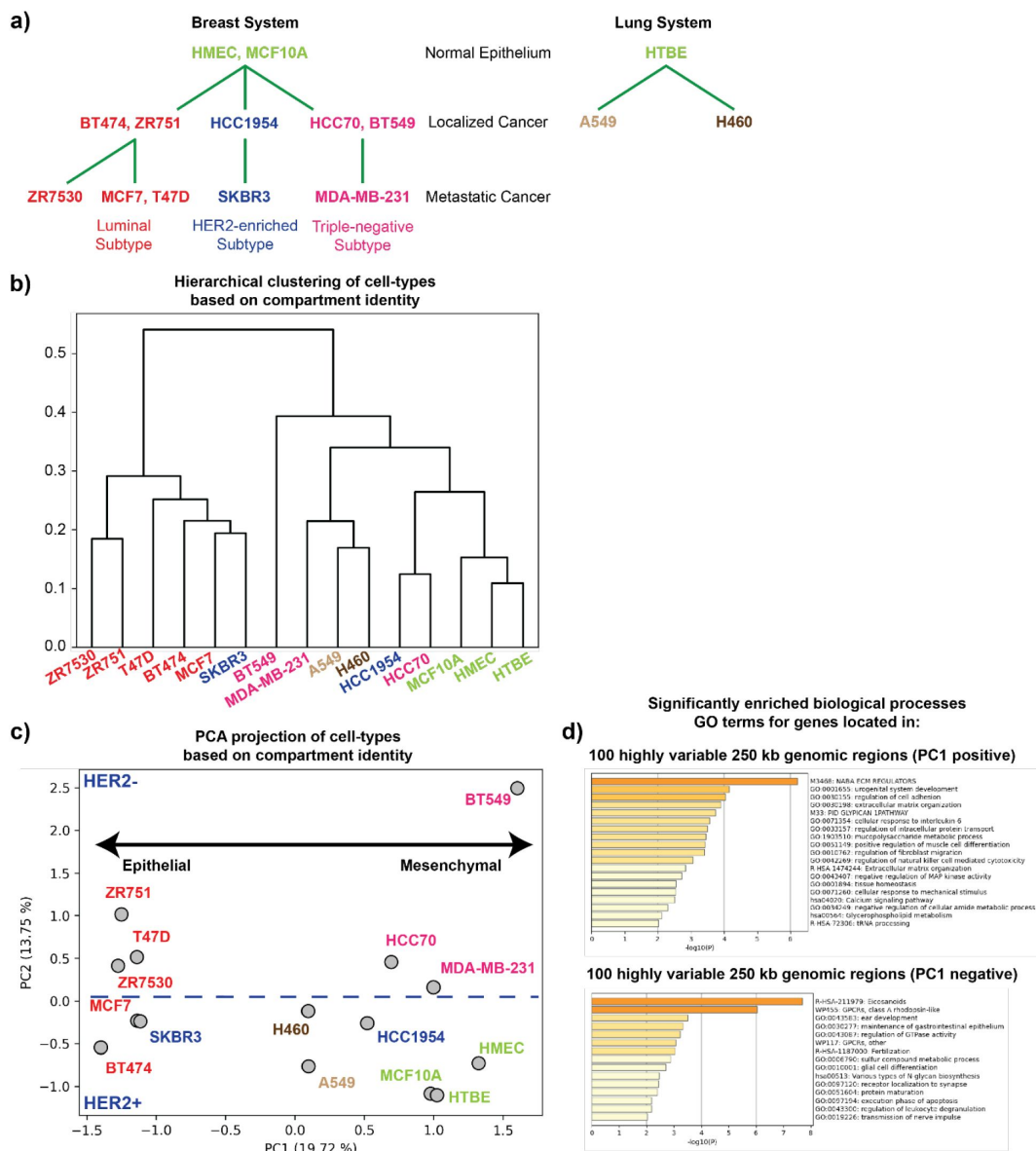


Fig 1.

Spatial compartment identity segregates cells by subtype and epithelial-mesenchymal status.

a) Primary cells and cell lines used in this study are arranged depending on their molecular subtypes and malignancy status based on prior available characteristics. Green: normal breast and lung epithelium, red: luminal subtype, blue: HER2-enriched subtype, pink: triple-negative subtype, brown: localized lung cancer. **b)** Hierarchical clustering of genome-wide compartment identity data for all breast and lung cell types at 250 kb resolution; colors as in **1a**. **c)** Principal component analysis of genome-wide compartment identity data for breast and lung cell types at 250 kb resolution; colors as in **1a**. The PC2 axis is divided into two subspaces that mostly segregate cell lines based on HER2 status (blue thick dotted line). **d)** GO term enrichment (biological processes) of the genes in genomic regions corresponding to the top 100 positive and 100 negative elements of the first eigenvector of genome-wide compartment identity PCA.

these cell lines based on transcriptomic data (Supplementary Figure 1) ([34](#), [35](#)). We noticed that the PC2 axis tends to segregate cell lines according to the presence or absence of HER2. As expected for an unsupervised analysis like PCA in a complex system, the separation is not perfect, as HER2 negative cell lines like MCF7 are plotted near HER2 positive cells. But, the trend of separation suggests that in breast cancer cell lines, the presence or absence of this receptor and its downstream signaling network has important effects on the genome organization at the compartment scale. To further characterize the genomic regions that showed distinctive compartment signatures across cell lines, we extracted lists of all genes that fell into genomic bins with the 200 most extreme eigenvector loadings for PC1: 100 strongest positive and 100 strongest negative bins. Gene ontology enrichment results for these gene lists coincide with the idea that PC1 represents the EMT axis: PC1 positive genes have functions related to cell adhesion (tenascin C, JAG1, ACTA2), extracellular matrix organization (APP, LAMA2, FGF2, BMP4) and positive regulation of cell migration, consistent with a mesenchymal phenotype, while PC1 negative genes show GO terms related to epithelial maturation (**Fig. 1d**). Overall, our analyses show that the genome organization is not only able to distinguish different subtypes of cancers but also exhibits changes along the epithelial-mesenchymal transition.

Breast cancer transcriptome profiles also exhibit epithelial-mesenchymal state-based segregation

We next analyzed the gene expression signatures that emerged across different subtypes of breast cancer (data sources described in Supplementary Table 1) and compared those with the patterns obtained from the compartmental analysis. We first performed PCA on the transcriptome profile of all the breast and lung system cells considered in this study (**Fig. 2a**). From the low dimensional PC projection result, we observed that similar breast cancer subtype cells distribute in the PC space based on their transcriptome profile. As in the compartment signature, MDA-MB-231 cells localize close to lung cancer cells A549 and H460, suggesting that this lung metastatic cell line has transcriptional similarities to lung cancer. We also noticed that along the largest PC axis (PC1), cells are ordered based on their epithelial-mesenchymal transition state. Performing GO term enrichment analysis using the genes from the regions with the top 100 most positive and 100 most negative eigenvector loadings for this PC1, we find that genes associated with PC1 positive values relate to epithelial organization, differentiation and negative regulation of cell migration, denoting a more epithelial phenotype (**Fig. 2b**). In contrast, PC1 negative genes relate to the mesenchymal functions with enriched terms for wound healing, hemostasis, cell movement, and ECM remodeling (**Fig. 2b**). Along with the unsupervised PCA, we also performed gene expression analysis in a supervised fashion using two different methods – gene set variation analysis (GSVA) and non-negative principal component analysis (nnPCA) (**Fig. 2c** and **2d**, see Methods).

Briefly, for each cell type, an E (epithelial) score and an M (mesenchymal) score were calculated based on the expression levels of a curated set of 232 epithelial and 193 mesenchymal genes ([35](#), [36](#)). A higher E score represents more epithelial characteristics whereas a higher M score corresponds to a mesenchymal phenotype. The GSVA and nnPCA transcriptome analyses recapitulate a similar ordering of cell lines according to epithelial-mesenchymal characteristics.

Distinct EMT gene sets are affected by spatial compartment changes vs. transcription changes, revealing multiple layers of EMT signature

Numerous studies have already demonstrated that cancer cells segregate along an EMT axis by transcriptomics ([34](#)–[39](#)). Given that spatial compartmentalization is often correlated with gene expression ([22](#)), a simple possible explanation for the EMT segregation of cells by compartment signatures could be that genes which change expression also change compartment. An alternative hypothesis would be that transcription and 3D genome organization change

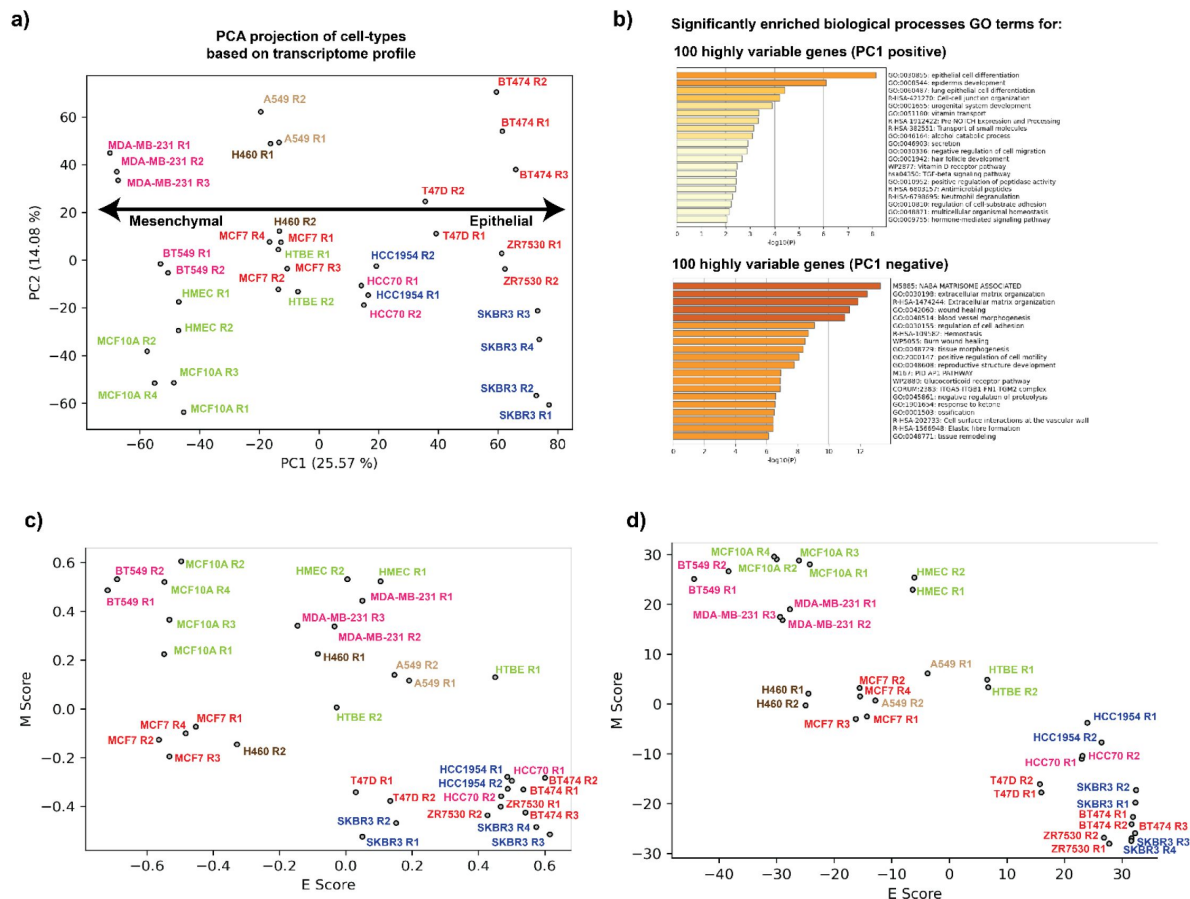


Fig 2.

Gene expression differences also segregate breast cancer cell lines along an EMT axis.

a) Principal component analysis of transcriptome data for breast and lung cell types in this study. Cell types are colored based on their molecular subtypes as represented in **Fig. 1a**. **b** GO term enrichment (biological processes) of the genes corresponding to top 100 positive and 100 negative elements of the first eigenvector of the transcriptome profile PCA. **c, d** Projection of breast and lung cell types on a curated epithelial and mesenchymal axis based on their gene expression using gene set variation analysis (GSVA) (**2c**) and non-negative principal component analysis (nnPCA) (**2d**) methods.

represent two independent aspects of the EMT transition. To evaluate these possibilities, we compared the genes obtained from the compartmental PCA analysis, the transcriptome PCA analysis, and the curated epithelial-mesenchymal gene set (**Fig. 3a**, gene lists found in Supplementary Table 2). We found that the genes in regions that change spatial compartmentalization have very little overlap with the sets of genes that most varied in transcription across these cell lines or were curated as differently expressed EMT genes (**Fig. 3a**). To further investigate this result, we examined the transcription status of genes that changed compartment across the EMT spectrum and, conversely, the compartment status of genes that changed transcription (**Fig. 3b, c**, and **d**). As expected, hierarchical clustering of the detected highly varying compartments (**Fig 3b**, top) and highly varying transcripts (**Fig 3c** and **d**, bottom) resulted in clear segregation of the cell types into similar groupings (i.e., HMEC and MCF10A together, BT474 and SKBR3 together with an opposing pattern). But, the genes which strongly changed compartments between cell lines showed no consistent patterns of correlated expression change (**Fig. 3b**, bottom). Likewise, very few compartment differences existed across cell lines at genes whose transcription segregated cells by EMT status (**Fig. 3c** and **d**, top). Notably, for both the transcriptome PCA analysis and the curated EMT gene set, the majority of the differentially expressed genes are found in the active A compartment. Overall, our results favor the hypothesis that compartmentalization and gene expression signatures are capturing two independent factors in EMT transition. The genes which differ most in expression across these cell lines are largely already accessible (A compartment) and therefore their expression is likely not controlled by spatial compartmentalization. Meanwhile, the additional EMT relevant genes that change compartment, but not transcription, may be now spatially poised for future gene regulation given additional environmental cues, as has been shown in other contexts (40–43). Overall, the set of genes that changes compartments does not have as clear epithelial and mesenchymal functional enrichment as the transcription change set of genes (as shown by the analysis of these gene sets presented in **Figure 1d** and **Figure 2b**). This could indicate that some of the compartment changes that occur with EMT are not directly gene regulatory but rather enable an overall conformational change of the chromatin that is needed for the alterations in physical cell state or to accomplish long distance gene regulation changes. Indeed, we observe specific cases where a compartment change region is located near but not at a gene expression change region (Supplementary Figure 1c). For example, TFF3 is in the set of genes that decreases in expression in M vs. E type breast cancer cell lines. While the gene itself remains in the A compartment in most all cell types, it lies near several regions that switch from A to B in M vs. E type cancer cell lines. Therefore, some of the discordant gene expression and compartment change regions may reflect the impact of compartment shifts on gene expression in neighboring regions. Notably, this same genomic region on chr21 showed striking compartment switches with cancer progression in prostate cancer as well.(19)

Lung metastatic breast cancer cell lines acquire lung-like genome architecture

To survive a metastatic environment, a cancer cell must express a repertoire of binding factors, extracellular matrix proteins, and ligands that positively interact with the secondary tissue microenvironment and foster colonization and growth. We hypothesized that for that to happen, the genome organization of the metastatic cell should change to resemble that of the cells at the secondary site, as a prelude of transcriptional activation. To evaluate this hypothesis, we compared the 3D genome organization of the metastatic and localized breast cancer cells with lung epithelial vs. lung cancer cells (**Fig. 4a, 4b**). As a baseline, we first compared the breast cancer compartment identity data with that of the normal breast epithelial cells. For each comparison, this resulted in four possible compartment identity combinations: AA (A compartment in both normal breast and breast cancer), AB (A compartment in normal breast and B compartment in breast cancer), BA (B compartment in normal breast and A compartment in breast cancer), and BB (B compartment in both normal breast and breast cancer). Similarly, the

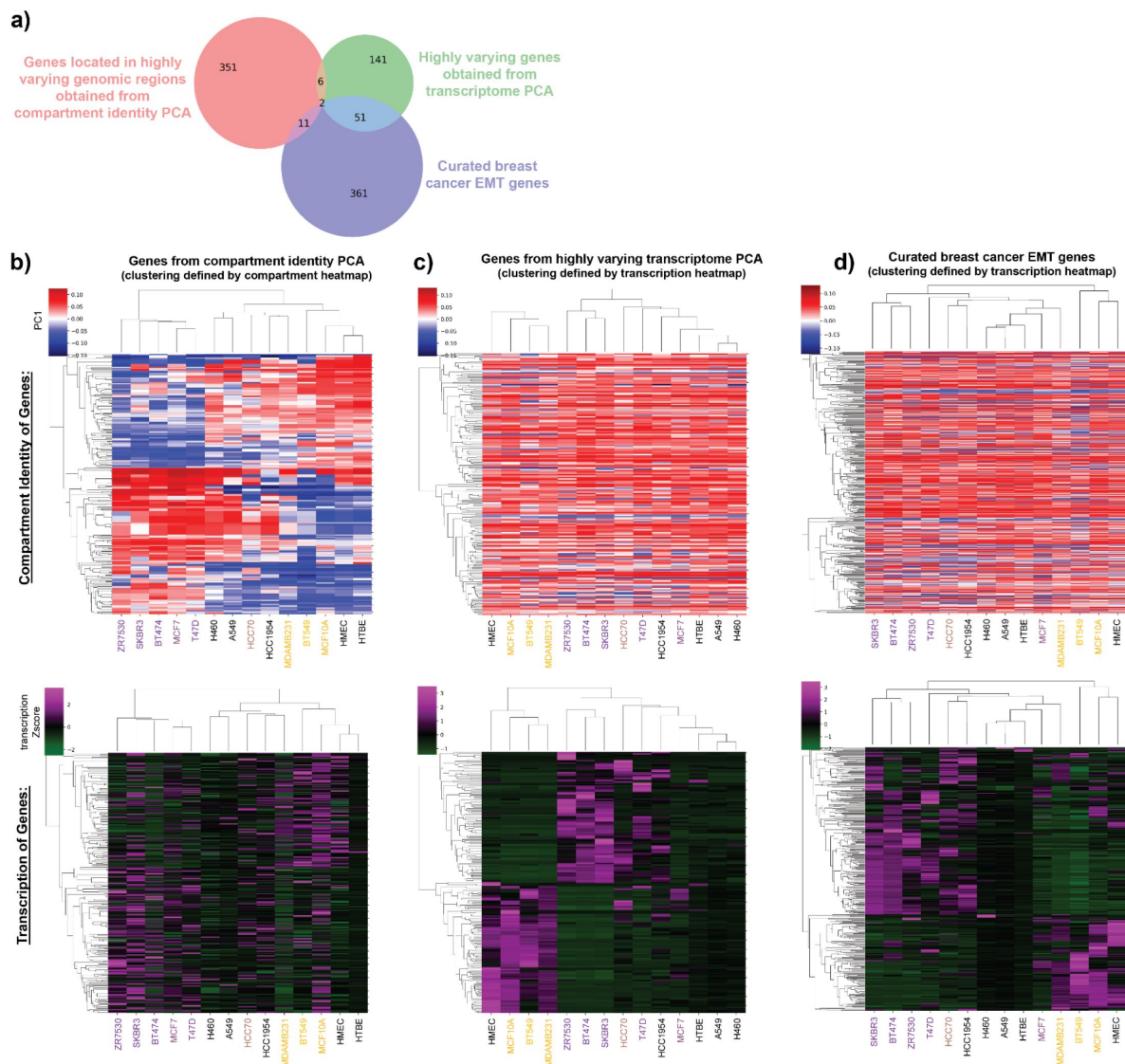


Fig 3.

Transcription and compartment changes capture distinct sets of EMT-related genomic regions.

a) Overlap analysis of the genes obtained from the compartmental analysis PC1 (red, same genes analyzed in [Figure 1d](#)), the transcriptome analysis PC1 (green, same genes analyzed in [Figure 2b](#)), and a curated breast cancer epithelial-mesenchymal gene set (blue). See Supplementary Table 2 for gene lists. **b-d)** Hierarchical clustering of compartment profiles (top row) and gene expression (bottom row) of the genes obtained from either **(b)** the compartmental analysis PC1, **(c)** transcriptome analysis PC1, or **(d)** the curated set of breast cancer epithelial and mesenchymal genes. For (b), the clustering order is determined by the compartment profile and then the same order of genes and cell lines is shown for the expression data. For (c) and (d), the clustering order is based on gene expression profiles and then the same order is shown for compartment profiles. Cell line names are colored according to position along the EMT axis (purple = more epithelial, yellow = more mesenchymal) as presented in Le et al., 2018 ([34](#)) and shown in Supplementary Figure 1a.

compartment identity of lung cancer cell lines was also compared with that of the normal lung cell. We then performed a cross-comparison between the breast and lung cancer comparisons, leading to a total of 16 compartment identity combinations (**Fig. 4b**). Of these, we were particularly interested in cases where the compartment state of healthy breast epithelial cells consistently differed from lung. It is notable that this set is relatively small to begin with (on the order of 200 250-kb bins), which emphasizes the overall similarities between epithelial cell types, whether they come from breast and lung. But for these regions that differ between healthy breast and lung, we next asked whether the breast cancer compartment would retain its breast identity or switch to match lung. In all the breast cancer cell lines derived from lung metastases, we found that more than half of these genomic regions switched compartments to match the lung compartment state (Supplementary Figure 2). We term these switches that make the breast cancer cell compartment identity more like the lung as “lung permissive genome organization changes”. They include: AB_BB (Genomic regions which are in the B compartment in both normal lung and localized lung cancer, and which change from A to B in the transition from normal breast to breast cancer) and BA_AA (Genomic regions which switch from B to A in breast cancer compared to normal breast epithelium, and which are in the A compartment in both normal lung and lung cancer). These combinations allow us to distinguish changes that could foster lung-specific-survival changes from cancer-specific traits such as AB_AB and BA_BA (compartments that change in the same direction from normal to cancer cells in both breast and lung). We then evaluated whether lung-permissive 3D genome changes occurred more frequently in lung metastatic breast cancer than in localized breast cancer.

To quantify the frequency of lung permissive changes overall, we calculated what fraction of the compartment alterations in breast cancer vs. normal breast tissue were also counted as lung permissive. We divided cells according to breast cancer subtype for this comparison. For the subtypes where several cell lines were available, we considered compartment identity combinations that were consistent between members of the group. We found that for luminal and triple-negative subtypes, metastatic breast cancer exhibits a higher level of lung permissive changes than localized cancer (**Fig. 4c**).

Genes affected by lung permissive compartment changes show lung-like expression levels cancer patients

After selecting the genomic regions that exhibit lung permissive changes in metastatic breast cancer cell lines, we extracted the underlying genes and used the GEPIA2 tool (44) to analyze the expression of these gene sets in TCGA breast and lung cancer patient data (**Fig. 4d**, 4e, and 4f; left panels). We found that in general, the gene expression levels change concordantly with the compartment. The results show that for the regions exhibiting lung-permissive changes, the switch in compartmental identity is associated with the gene expression change from normal to tumor. That is, for BA_AA regions, the breast tumor expression of these genes is higher than normal breast tissue expression, coinciding with higher expression in the lung (**Fig. 4d**). Conversely, for AB_BB regions, the breast tumor expression of genes in these regions is lower than for normal tissue, corresponding to lower expression in the lung (**Fig. 4e**). Notably, these lung-permissive compartment change genomic regions do not strongly overlap with the key compartment switch regions we identified by principal components analysis (Supplementary Table 3). Correspondingly, the enriched functions of genes in these lung permissive switch regions (**Figure 4d-f**) are different than the EMT-related functions found among the genes identified by PC1.

Organotropic 3D genome changes match target organ more than an unrelated organ

To verify that “lung permissive changes” truly occurred more than expected in lung metastatic cells, we used a different organ system-brain metastasis and brain cancer-as a negative control. Breast cancer can metastasize to brain(45,46), but the specific metastatic cell lines for which

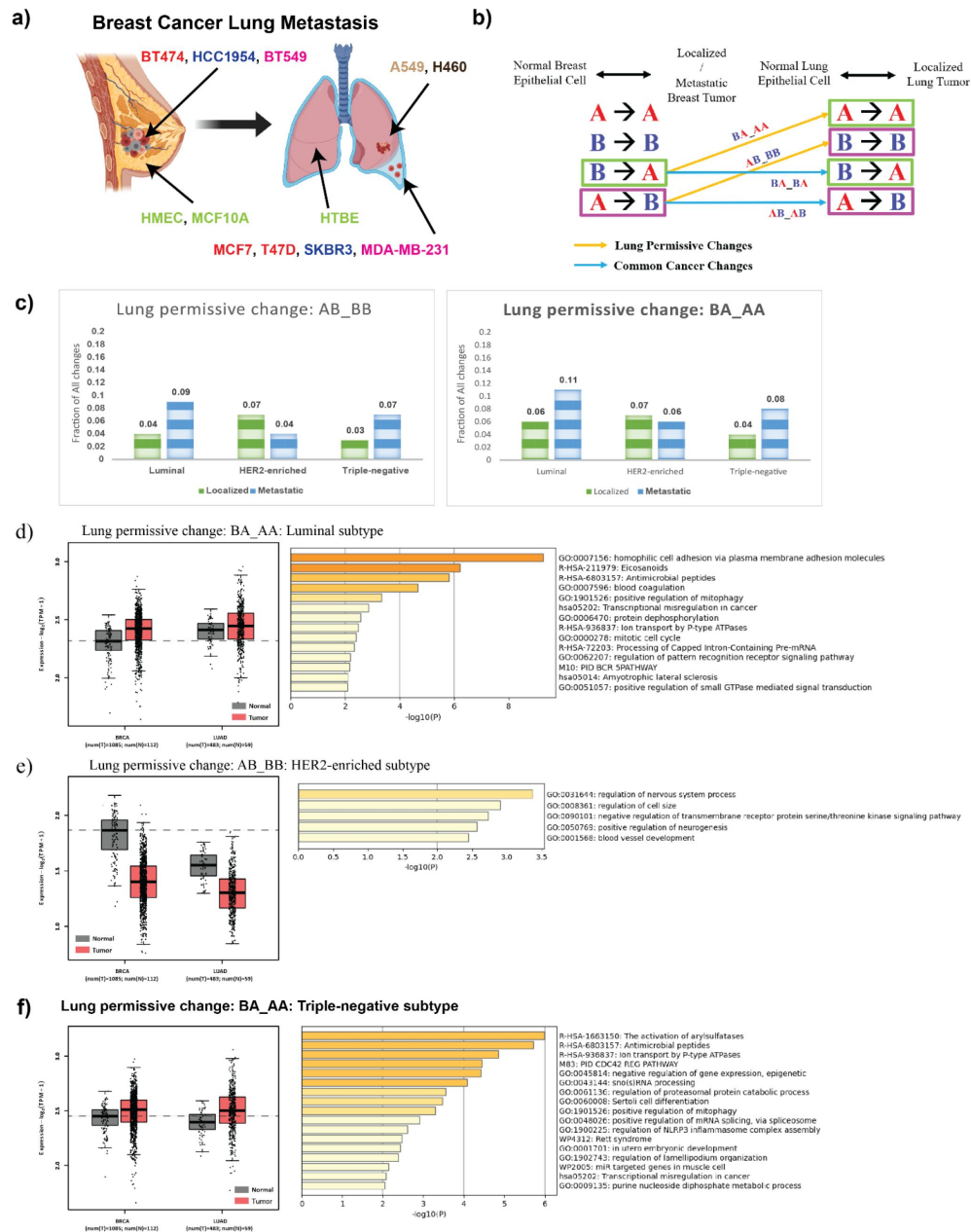


Fig 4.

a) Normal breast epithelium (HMEC and MCF10A), localized breast cancer (BT474, HCC1954, and BT549), and lung metastatic breast cancer (MCF7, T47D, SKBR3, and MDA-MB-231) cells are used to model different stages of breast cancer progression. To model diseased lung, normal lung epithelium (HTBE) and localized lung cancer (A549 and H460) cells are used. **b)** Schematic diagram representing breast cancer lung permissive changes calculation. First, for both breast and lung, cancer cell compartment identity is compared with the normal epithelial cell. This results in four possible compartment identity combinations: AA (A compartment in both normal and cancer), AB (A compartment in normal and B compartment in cancer), etc. Then, a cross-comparison between the breast and lung systems leads to 16 compartment identity combinations. Two specific combinations: AB_BB and BA_AA are defined as lung permissive changes (yellow arrows). **c)** Fraction of lung permissive changes shown by localized and metastatic cancers from different breast cancer subtypes. **d-f)** TCGA patient gene expression (left) and GO term enrichment (biological processes) (right) for genes from regions exhibiting **(d)** BA_AA lung permissive changes in the luminal metastatic breast cancer subtype, **(e)** AB_BB lung permissive changes in case of HER2-enriched metastatic breast cancer subtype, or **(f)** BA_AA lung permissive changes in case of triple-negative metastatic breast cancer subtype. TCGA data from BRCA and LUAD tumor vs. normal sets used and sample size indicated.

we have data were not derived from brain metastases. Therefore, we would expect fewer “brain like” compartment changes in metastatic breast cancer that metastasizes to lung. Before comparing percent permissive changes between organs, we developed an adjusted version of the calculation to account for differing baseline probabilities of compartment change at the distant organ site (Supplementary Figure 3 and Methods). The adjusted calculation still results in a higher level of lung permissive changes in lung metastatic breast cancer cell lines as compared to localized (Supplementary Figure 4d). For the brain comparison, we used the NHA cell line (normal human glia) and the SF9427 cell line (glioblastoma) as our reference models (Supplementary Figure 4a and 4b). We then calculated the adjusted level of brain permissive changes shown by localized and metastatic breast cancers (Supplementary Figure 4c). We found that the metastatic breast cancer cells used in this study showed lung-permissive changes more frequently than brain-permissive compartment changes (Supplementary Figure 4c and d). This is congruent with the fact that these metastatic breast cancer cell lines were all isolated from lung pleural effusions. We note that Hi-C data for brain metastatic breast cancer cell lines is not currently available for comparison.

Brain metastatic prostate cancer cell lines exhibit organotropic 3D genome changes toward brain identity

The analysis above suggests that lung metastatic breast cancer cell lines show more lung-permissive than brain-permissive genome organization changes. Next, we asked whether a cancer type that metastasizes to the brain exhibits brain-permissive compartment changes. For this purpose, we chose prostate cancer cell lines derived from localized (lymph node) and brain metastatic sites (LNCaP and DU145) and compared their genome architecture to the non-tumorigenic prostate epithelial cell line RWPE (Supplementary Figure 5a, b). We found that the brain metastatic cells showed more brain-permissive changes than the one derived from a lymph node (Supplementary Figure 5c). Also, particularly in the B to A direction, there were greater brain permissive changes in this prostate cancer system (Supplementary Figure 5c) than in the breast cancer system (Supplementary Figure 4c). This confirms that a cancer type that metastasized to the brain matched the brain genome compartmentalization profile more often than a cancer type that had not metastasized to the brain. We found that the genes in the switched compartment regions have functions that could relate to leaving the prostate epithelium and adapting to the brain environment (Supplementary Figure 5d). For example, the DU145 (brain metastatic) brain-permissive B to A switched genes were enriched for functions in regulation of synapse organization (including calcium channel CACNA2D3). Conversely, the regions that switch from A to B to match the brain B compartment and are enriched for keratinization and cell adhesion, suggesting a downregulation of prostatic epithelial properties.

Discussion

During metastasis, breast cancer cells spread preferentially to certain organs, a series of events widely known as metastasis organotropism. This phenomenon, also shown by other types of cancer, may be explained, to some extent, by the ‘seed and soil’ hypothesis which states that cancer cells will be able to colonize a distant site if the cells have intrinsic ability to survive in the microenvironment of the secondary site. Though the process of metastasis is complex and involves other factors, the primary tumor microenvironment conveys survival capabilities to cancer cells that can potentially favor survival in other organs; for this to occur, the cancer cell must engage specific gene expression signatures. However, there are many open questions related to the rise of organ-specific survival traits. Are these gene signatures a result or consequence of genome organization changes? What is the role of genome rearrangements in metastasis organotropism for which there are no significant gene expression alteration? Does the genome organization of

the metastatic cell become similar to that of the cells at the distant site? In this work, we have made a step toward answering these questions by analyzing compartmental organization data of breast cancer cell lines derived from lung metastasis in comparison to lung cell lines.

Unsupervised clustering of compartment organization data showed that compartment organization can distinguish cancer cell lines by subtype, providing the first indication that lung metastatic cell lines may show similarities to lung cell genome organization. For example, the triple-negative invasive metastatic breast cancer cell MDA-MB-231, clusters close to the lung carcinoma cells (A549 and H460). The clustering result supports the conclusion that the breast cancer subtypes are not only quite different at the gene expression level, but also at the level of genome organization. Furthermore, it also shows that the genome organization is highly consistent among luminal subtype cancers, irrespective of the tissue from which the cells were collected, while the other subtypes such as triple negative, were more heterogeneous in their structure. This is congruent with studies involving gene expression which have shown that the triple-negative breast cancer category is highly heterogeneous, encompassing six subtypes (47 [↗](#)).

When we compared the compartmental organization changes shown by the localized and lung metastatic breast cancers to the compartment state of the lung, we found that, for luminal and triple-negative subtypes, metastatic cells exhibit higher percentages of lung permissive changes compared to localized cancers. Given that these cancer cell lines were derived from lung pleural effusions, we cannot tell whether these metastatic cells acquired a lung-like compartmental organization at the primary tumor site or later while adapting to the metastatic site. To answer this, we would need temporal genome organization data of metastatic progression, which is not currently available. Our current study is also limited by the use of cell lines derived from multiple different patient sources and grown in culture. To validate and clarify our observations, future research would need to analyze the 3D genome structure of matched primary and metastatic tumors from the same human patient.

For the lung permissive chromatin conformation changes, we find that shifts toward the A compartment are favored compared to the B compartment. It may be possible that shifts toward the A compartment are predominant because both activation and gene repression are possible once a region is in the A compartment while a shift to the B compartment is not necessary to accomplish gene repression (40 [↗](#), 48 [↗](#), 49 [↗](#)). Alternatively, this preference for shifts toward an A compartment could also be related to the general loss of heterochromatin previously observed in cancer (50 [↗](#)). Indeed, we found that genes in these spatially reorganized regions, in patient samples, tend to change expression concordantly with their compartment changes. Overall, our results are consistent with the idea that genome compartmentalization can play a role in facilitating organ-specific metastasis.

While our gene expression and compartment analyses echo the current dogma that 3D genome compartments correlate with gene expression changes, further analysis revealed that gene expression and compartment signatures across breast cancer types surprisingly capture two orthogonal sets of genomic regions. We found that compartment profiles group breast cancer cell lines according to their epithelial vs. mesenchymal states (51 [↗](#)). This observation becomes much more visible in the PCA transformed data as the cells are arranged along an epithelial-mesenchymal spectrum. Pathways of EMT such as the loss of epithelial identity and the gain of migratory ability can play important roles in facilitating metastasis, though it is important to note that EMT is only one of many complex processes involved in metastasis. (52 [↗](#)) While epithelial vs. mesenchymal states have been clearly associated with gene expression, it is also becoming clear that these states exhibit characteristic genome architectures as well. Indeed, a recent study shows that genome organization changes during epithelial-mesenchymal transition in ovarian cancer system, by considering some representative epithelial and mesenchymal cell lines (53 [↗](#)). Here, we extend this conclusion for another major cancer, the breast cancer system by including a spectrum of breast and lung cancer and normal cell lines. Across this cohort, we find not only that both 3D

genome architecture and gene expression changes occur during EMT, but also that these changes happen in different regions. The 3D genome changes that most strongly capture the differences across the epithelial and mesenchymal spectrum occur at regions where gene expression is not notably distinct between epithelial and mesenchymal cells. Likewise, the signature gene expression changes occur in regions that tend to be in the A compartment across all cell types. This suggests that many genes key to EMT are regulated by local factors in the A compartment, while major spatial shifts in chromosome regions may alter long-distance regulation of genes in other regions, change the likelihood of future inducibility of genes, or represent alterations in the physical/mechanical properties of the cell during EMT, which can affect chromatin state (54 [↗](#), 55 [↗](#)). Overall, our results suggest that genome spatial compartment changes can help encode cancer cell states that may play a role in metastasis as well as enabling survival in a new organ context.

Methods

Hi-C data processing

To process Hi-C sequencing reads, we used the HiC-Pro (<https://github.com/nservant/HiC-Pro> [↗](#)) pipeline with default parameters (56 [↗](#)). The reads were aligned to the hg19 reference genome. Also, for each cell type, all the available reads from all replicates were merged together to improve the quality of contact data for further downstream processing. After obtaining the valid pair interactions, we binned the interactions at 250 kb resolution for each individual chromosome and normalized the matrices using the ICE technique (57 [↗](#)). Following that, the number of normalized contacts of each chromosome was scaled to 1M using 'scaleMatrix.pl' script from the cworld-dekker (<https://github.com/dekkerlab/cworld-dekker> [↗](#)) tool suite. And finally, in order to get the compartmental organization for each individual chromosome, principal components analysis was performed on the normalized and scaled matrices using the 'matrix2compartment.pl' script from the cworld-dekker tool suite. The compartment analysis assigns A/B compartment identity to genomic regions based on the sign of the elements of the eigenvector (A-positive and B-negative).

RNA-seq data processing

The RNA-seq files were processed as per the steps described in the previous publication (40 [↗](#)). Briefly, the BBDuk (<https://github.com/kbaseapps/BBTools> [↗](#)) tool was used first for adapter and quality trimming (58 [↗](#)). After that, the reads were mapped to hg19 human reference genome using STAR aligner (RRID:SCR_004463) (59 [↗](#)). Once aligned reads were generated, HTSeq-Counts (<https://github.com/simon-anders/htseq> [↗](#)) tool was used to perform gene level feature count (60 [↗](#)). As the RNA-seq files were obtained from various sources and generated using different experimental techniques, they suffer from the batch effect which masks the biological signal. To remove the batch effect, ComBat-seq (<https://github.com/zhangyuqing/ComBat-seq> [↗](#)) tool was applied on the raw gene counts data (61 [↗](#)).

Hierarchical clustering of genome-wide compartmental organization

To cluster different cell types based on their compartmental organization, hierarchical clustering was performed on the genome-wide compartment identity data. First, the pairwise similarity score between all the cells were calculated using Pearson's correlation. The similarity data was then further converted to distance data by subtracting from 1. And finally, hierarchical clustering was performed on the distance data using 'ward' linkage.

Gene set variation analysis (GSVA)

GSVA was performed to compute the enrichment scores for individual cell lines using RNA-seq data (62,63). A list of 232 epithelial signature genes, and a list of 193 mesenchymal signature genes were used to compute the E enrichment score (E score) and M enrichment score (M score) respectively (35,36,64).

Non-negative principal component analysis (nnPCA)

We performed nnPCA using the gene sets mentioned above. nnPCA provided higher resolutions for large number of samples in the EMT spectrum compared to GSVA according to previous studies (36,65). nnPCA determines the approximately orthogonal axes with non-negative coefficients (loadings) for features (genes). Variances of projections of data points (cell lines) onto these axes are maximized via an optimization method (66).

Correction for comparing organ permissive changes across different organs

It is possible that while comparing the host organ permissive changes across different secondary organs, the compartment switch ratio of different secondary localized cancers might include inherent bias to the comparison. To eliminate the bias, for the secondary organ systems, we first calculate the probability of a genomic region to be in a specific compartment in localized cancer given that the region belongs to the same compartment in normal cell (Supplementary Figure 3). For the A compartment, $p(TA|NA)$ represents the probability of a region in A compartment in localized cancer cell given that region also belongs to A compartment in normal cell. Similarly, $p(TB|NB)$ for the B compartment. If we have more than one cell line representing the secondary site localized cancer, the minimum probability across all the cells is considered. Once we have those, the probabilities are subtracted from one and the resultant values are multiplied with corresponding secondary organ specific percentages. For example, $(1-p(TA|NA))$ is multiplied to the percentage of BA_AA and $(1-p(TB|NB))$ to the AB_BB.

Data availability

Datasets used for this project were obtained from NCBI GEO, EMBL ENA, 4DN Data Portal and ENCODE repositories and their accession IDs are given in Supplementary Tables 1 and 2.

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Additional information

Author contributions

P.D. conceived the study in collaboration with R.P.M and R.S.M., performed data analyses, and prepared figures. R.S.M. performed the grouping and ordering of cancer cell lines into biological subtypes and interpreted the biological enrichment analysis. T.H. performed EMT gene signature analyses. R.P.M. supervised and guided the work. P.D. and R.P.M. wrote the manuscript with input from all authors.

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Additional files

Supplementary Figures and Tables [↗](#)

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Editors

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Reviewer #1 (Public review):

Summary:

This study utilized publicly available Hi-C data to ensemble a comprehensive set of breast cancer cell lines (luminal, Her2+, TNBC) with varying metastatic features to answer whether breast cancer cells would acquire organ-specific feature at the 3D genome level to metastasize to that specific organ. The authors focused on lung metastasis and included several controls as the comparison including normal mammary lines, normal lung epithelial lines, and lung cancer cell lines. Due to the lower resolution at 250KB binning size, the authors only addressed the compartments (A for active compartment and B for inactive compartment) not the other 3D organization of the genome. They started by performing

clustering and PCA analysis for the compartment identity and discovered that this panel of cell lines could be well separated based on Her2 and epithelial-mesenchymal features according to the compartment identity. While correlating with the transcriptomic changes, the authors noticed the existence of concordance and divergence between the compartment changes and transcriptomic changes. The authors then switched gear to tackle the core question in metastatic organotropism to the lung. They discovered a set of "lung permissive compartment changes" and concluded that "lung metastatic breast cancer cell lines acquire lung-like genome architecture" and "organotropic 3D genome changes match target organ more than an unrelated organ". To prove the latter point, the authors enlisted additional non-breast cancer cell line (prostate cancer) in the setting of brain metastasis. This is a piece of pure dry computational work without wet bench experiments.

Strengths:

The authors embarked on an ambitious journey to seek for the answer regarding 3D genome changes predisposing metastatic organotropism. The authors succeeded in the assembly of a comprehensive panel of breast cancer cell lines and the aggregation of the 3D genome structure data to conduct a hypothesis driven computation analysis. The authors also achieved in including proper controls representing normal non-cancerous epithelium and the end organ of interest. The authors did well in the citation of relevant references in 3D genome organization and EMT.

<https://doi.org/10.7554/eLife.103697.2.sa2>

Reviewer #2 (Public review):

Summary:

This work addresses an important question of chromosome architecture changes associated with organotopic metastatic traits, showing important trends in genome reorganization. The most important observation is that 3D genome changes consistent with adaptations for new microenvironment, including lung metastatic breast cells exhibiting signatures of the genome architecture typical to a lung cell-like conformation and brain metastatic prostate cancer cells showing compartment shifts toward a brain-like state.

Strengths:

This work presents interesting original results, which will be important for future studies and biomedical implications of epigenetic regulation in norm and pathology.

<https://doi.org/10.7554/eLife.103697.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Strengths:

The authors embarked on an ambitious journey to seek the answer regarding 3D genome changes predisposing to metastatic organotropism. The authors succeeded in the assembly of a comprehensive panel of breast cancer cell lines and the aggregation of

the 3D genome structure data to conduct a hypothesis-driven computation analysis. The authors also achieved in including proper controls representing normal non-cancerous epithelium and the end organ of interest. The authors did well in the citation of relevant references in 3D genome organization and EMT.

Weaknesses:

(1) The authors should clearly indicate how they determine the patterns of spread of the breast cancer cell lines being utilized in this manuscript. How did the authors arrive at the conclusion that certain cell lines would be determined as "localized spread" and "metastatic tropism to the lung"? This definition is crucial, and I will explain why.

It is indeed a critical point to clearly define and explain what qualifies as metastatic potential to particular organs in our system. Here, we intentionally limited our scope to metastasis that had occurred within the human system. Our cell lines are chosen based on their sites of origin and etiological history in the patients from which they were derived. For example, the cancer cell line BT474 was classified as “localized” because these cells were derived from a solid tumor in the breast itself. Meanwhile, MCF7 and T47D cell lines are considered lung metastatic because these cells were collected from the pleural effusion from the lung. We therefore model human organotropism from the breast to the lung by using cells that originated from infiltrative ductal carcinoma (human breast) but were collected from pleural effusions (human lung). We then use as a comparison a human lung cancer-derived cell line that was itself purified from a pleural effusion. In this way, we can compare the genome structure of a lung cancer cell in the lung environment to a breast cancer cell that has metastasized to the lung environment.

In our revised version, we further clarify this definition in the text as well as in additional annotations in our supplemental table of all cell line information.

Todd Golub's team from the Broad Institute of MIT and Harvard published "A metastasis map of human cancer cell lines" to exhaustively create a first-generation metastasis map (MetMap) that reveals organspecific patterns of metastasis. (By the way, this work was not cited in the reference in this manuscript.) The MetMap Explorer (<https://depmap.org/metmap/vis-app/index.html>) is a public resource that could be openly accessed to visualize the metastatic potential of each cell line as determined by the in vivo barcoding approach as described in the MetMap paper in the format of petal plots. 5 organs were tested in the MetMap paper, including brain, lung, liver, kidney, and bone. The authors would discover that some of the organ-specific metastasis patterns defined in the MetMap Explorer would be different from the authors' classification. For example, the authors defined MCF7 as a line as lung metastatic, and rightly so the MetMap charted a signal towards lung with low penetrance and low metastatic potential. The authors defined ZR751 as a line with localized spread, however, the MetMap charted a signal towards the kidney with low penetrance and low metastatic potential, the signal strength similar to the lung metastasis in MCF7. A similar argument could be made for T47D. The TNBC line MDA-MB-231 is indeed highly metastatic, however, in MetMap data, its metastasis is not only specific to the lung but towards all 5 organs with high penetrance and metastatic potential. The 2 lung cancer cell lines mentioned in this study, A549 and H460, the authors defined them as localized spread to the lung. However, the MetMap data clearly indicated that A549 and H460 are highly metastatic to all 5 organs with high penetrance and high metastatic potential.

We acknowledge the valuable contributions of animal models in metastatic cancer studies, but we also want to avoid the potentially confounding variable of the animal microenvironment. The MetMap Explorer contains valuable information (and as part of our clarification on this point, we now cite the MetMap in the text), but the “metastatic potential

of each cell line” for this tool is measured in a mouse environment. Knowing that a particular cell line, which originated from a human lung metastasis, can further metastasize to other organs in a mouse does not necessarily mean that those cells could do so in humans. The microenvironment responses to metastatic colonization recapitulate the events in wound repair, and these can differ among species (<https://pubmed.ncbi.nlm.nih.gov/28916657/><https://pubmed.ncbi.nlm.nih.gov/39729995/>). Further, the changes a cell needs to make to adapt to a new organ system in a mouse could be confounded by the changes needed to adapt to mouse conditions in general. Finally, migration from a site of ectopic injection may not mimic migration from an initial tumor site. These factors lead to well known cases where MetMap does not reflect the metastatic potential of cancers in humans. As a classic example, prostate cancer frequently metastasizes to bone in humans, and the PC3 cell line was derived from a bone metastatic prostate cancer. However, MetMap shows no evidence of PC3 being able to metastasize to bone in a mouse.

We agree that the very best data would come from matched primary and metastatic tumors in the same human patient, but those data do not currently exist and generating them would require future work beyond the scope of this study.

Since results will vary among different experimental models testing metastatic organotropism, (intracardiac injection was the metastasis model being adopted in the MetMap), the authors should state more clearly which experimental model system served as the basis for their definition of organ-specific metastasis. In my opinion, this is the most crucial first step for this entire study to be sound and solid.

Taking all the above into account, in our revision, we have now included further clarification in the main text to more clearly explain how and why we chose the cell lines we did and what the advantages and limitations of this choice are.

(2) Figure 1b: The authors found that "MDA-MB-231 cells were grouped with the lung carcinoma cells. This implies that the genome organization of this cell line is closer to that of lung cells than to other breast epithelial cell lines." In fact, another TNBC line BT549 was also clustered under the same clade. So this clade consisted of normal-like and highly metastatic lines. Therefore, the authors should be mindful of the fact that the compartment features might not directly link to metastasis (or even metastatic organotropism).

In figure 1b, the grouping that includes MDA-MB-231 (lung metastatic breast cancer) connected to A549, and H460 (lung cancer) occurs at a distance of about 0.2. If the clustering tree were cut at a distance of 0.26, 6 separate clusters would result: two clusters of Luminal subtypes (all labeled red), one that includes all healthy epithelial cells (both lung and breast, all labeled green), one that links two localized breast cancers, one that links MDA-MB-231 to lung carcinoma cell lines, and then BT549 by itself. So, while BT549 appears next to MDA-MB-231 along the horizontal axis, this is just coincidence of the representation: the dendrogram shows it is quite distant from all the other cell lines in this cluster according to compartment profile.

So, it is only MDA-MB-231 that is very closely linked with the lung cancer cell types.

It is true that the healthy lung cells (HTBE) are clustered separately and are more similar to normal/non tumorigenic breast epithelial cells (HMEC and MCF10A) than to any cancer cell type. This could suggest that there are aspects of the compartment pattern that represent any healthy epithelium as compared to cancer. What we find in the compartment profile, in both the clustering and the PCA analysis, is that compartment signatures contain information about cell properties on several overlapping levels: there is an aspect of the compartment profile that distinguishes healthy from cancerous cells, an aspect that distinguishes luminal

cancers from other subtypes, a part that associates with organotropism, and an aspect that captures EMT status. The final compartment status is a composite of these numerous factors.

We have clarified the text to indicate that we mean MDA-MB-231 clusters near lung cancer, not necessarily healthy lung cell models.

(3) Figure 3: In the text, the authors stated, "To further investigate this result, we examined the transcription status of genes that changed compartment across the EMT spectrum and, conversely, the compartment status of genes that changed transcription (Fig. 3b, c, and d)". However, it was not apparent in the figure that the cell lines were arranged according to an EMT spectrum.

To display these comparisons more clearly, we have now revised figure 3b, c, and d in two ways: First, we have defined the gene and cell line clustering by one set of data (for example, compartment identity in 3b) and then displayed the other data (gene expression) with all genes and cell lines in the same order. Therefore, for each column, genes and cell lines can be compared visually between top and bottom rows. Second, we have colored cell line names from purple to yellow according to their EMT scores as shown in Supplementary Figure 1a. This allows a visual indication of how the clustering separates cell lines by EMT status.

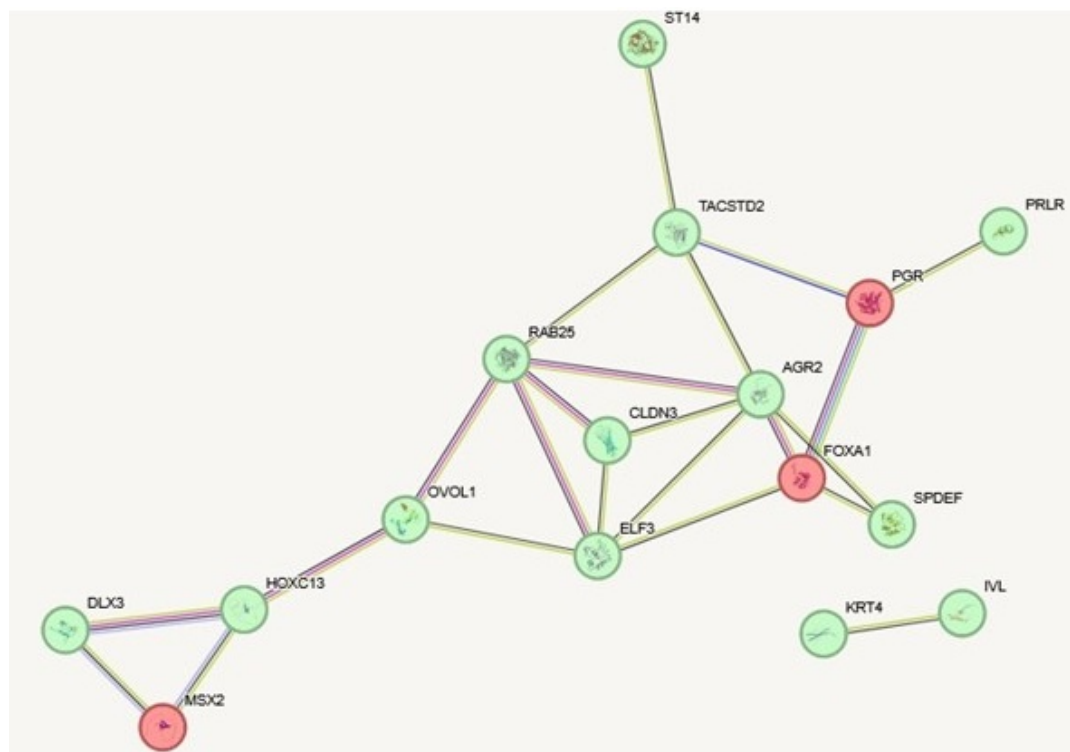
Also, the clustering heatmaps did not provide sufficient information regarding the genes with concordant/divergent compartments vs transcription changes. It would be more informative if the authors could spend more effort in annotating these genes/pathways.

We want to clarify that the genes plotted in the heatmaps in Figure 3 are also the genes whose functional enrichment we present in figures 1 and 2. So, the genes that segregate strongly based on A/B compartment (but not gene expression) in figure 3b are the same genes whose GO terms are annotated in Figure 1d. Likewise, the genes that segregate strongly based on gene expression, but not A/B compartment, in figure 3c and d are the same genes whose GO terms are annotated in Figure 2b. We have now made this connection clearer in the text.

But, we also agree with the reviewer that it is important to explore a bit further the relationship between these divergent sets of genes. Our explorations have led to several observations:

(1) In some cases, the compartment-segregated genes and the transcription-segregated genes are different members of the same pathways. In Author response image 1 below, for example, we show interactions (according to STRING) for genes from figure 3c that are highly expressed in the epithelial-like cell lines and are annotated as involved in epithelial development (green). We then added to the network genes from figure 3b that are specifically in the A compartment in the epithelial-like cell lines but not mesenchymal cell lines that are also annotated as involved in epithelial development (red). Most of these epithelial development genes that change expression are in the A compartment in all cell lines and therefore do not rely on spatial compartment changes for their regulation. But some additional epithelial development genes, which are interconnected in this same network, are changing compartments across the EMT spectrum. One example, FOXA1, is a key hub in the network and is known to be a pioneer transcription factor involved in development and differentiation. Controlling this gene at the level of spatial genome organization rather than local transcriptional control could be important in the stable cell fate changes that can happen with EMT.

Author response image 1.



(2) Overall, the set of genes that change compartments does not have as strong functional enrichment as the transcription change set of genes. This could indicate that some of the compartment changes that occur with EMT are not directly gene regulatory but rather enable an overall conformational change of the chromatin that is needed for the alterations in physical cell state or to accomplish long distance gene regulation changes.

(3) Related to long distance gene regulation changes, we also see cases in which the gene that changes transcription but not compartment across EMT is adjacent to regions that switch compartments.

A good example is TFF3 (yellow, Supplementary figure 1C). TFF3 is one of the genes that strongly segregates across EMT by transcription, being more highly expressed in epithelial-like (bottom 4 tracks) but not mesenchymal-like (top 4 tracks) cancers. Despite this differential expression, it is almost always in the A compartment across all cell lines. However, it is adjacent to regions that show strong compartment change EMT signatures. So, even though this specific gene region is not changing compartment, its regulation may be influenced by the entire region being A-associated in epithelial-like but neighboring regions becoming B-associated in mesenchymal like cancers.

TFF3 is expressed in normal breast epithelium and has been implicated as a biomarker for endocrine therapy response in breast cancer.

Meanwhile, many genes that are in these compartment switching regions (BACE2, DSCAM, PDE9A) are not among the strongest expression signature genes.

(4) Interestingly, some of the regions (such as the region shown in Supplementary figure 1C) that change compartment across the breast cancer spectrum overlap with regions that we found change compartment in the progression of prostate cancer, as shown in the string.db enrichment analysis below.

Author response image 2.

Functional enrichments in your network [explain columns](#)

Reference Publications (PubMed)					
publication	(year) title	count in network	strength	signal	false discovery rate
PMID:19783846	(2009) A new mouse model for the trisomy of the Abcg1-U2af1 reg...	12 of 37	1.69	2.3	1.43e-09
PMID:34889941	(2022) Chromosome compartmentalization alterations in prostate ...	10 of 49	1.49	1.29	1.26e-05
PMID:26765563	(2016) Genetic dissection of Down syndrome-associated congenit...	8 of 25	1.69	1.16	5.63e-05
PMID:35331298	(2022) Familial microdeletion 18p11.32 to 18p11.31 in a Chinese f...	7 of 17	1.8	1.07	0.00014
PMID:36995257	(2023) Overexpression screen of chromosome 21 genes reveals m...	11 of 93	1.25	1.01	0.00011

[\(more ...\)](#)

In our revised manuscript, we now include more of these explanations in the text and include the example offset compartment and transcription change region shown about as panel c of Supplementary Figure 1.

(4) Figure 4: The title of the subheading of this section was 'Lung metastatic breast cancer cell lines acquire lung-like genome architecture'. Echoing my comments in point 1, I am a bit hesitant to term it as "lung metastatic" but rather "metastatic" in general since cell lines such as MDA-MD-231 do metastasize to other organs as well. However, I do get the point that the definition of "lung metastasis" is derived from the common metastasis features among the cell lines here (MCF7, T47D, SKBR3, MDAMB-231). There might be another argument about whether the "lung" carcinoma cell lines can be considered "localized" since they are also capable of metastasizing to other organs.

Rather than classifying cells on metastatic “potential” (as measured in a mouse), our cell lines are chosen based on their sites of origin and etiological history in the patients from which they were derived. Cancer cell lines called “lung metastasis” were collected from the pleural effusion from the human lung. Likewise, we call a cancer “localized” because it was taken from the tissue where the cancer originated, even if it might, if placed into a different context, be able to metastasize. We would argue that the genome structure features of the “localized” cancers reflect cancers that have not yet metastasized (even if they could in the future) while the “metastatic” cancers have already gone to a certain location (even if they could in theory have gone to a different location).

In a way, what the authors probably were trying to leverage here is the "tissue" identity of that organ.

Having said this, in addition to showing the "lung permissive changes", the authors should show the "breast identity conservation" as well. Because this section started to deal with the concept of "tissue/lineage identify", the authors should also clarify whether these breast cancer cell lines capable of making lung metastasis are also preserving their original tissue identity from the compartment features (which would most likely be the case).

This is a great question. We have now more explicitly checked the proportions of genomic regions that change compartments to match lung vs. maintaining breast-specific compartment identity. The graphs in Supplementary Figure 2 begin with all genomic bins that have distinctive compartment identity between non-cancerous breast and lung epithelial cells. Then, the plots show what fraction of these tissue-specific bins change compartment to match lung vs. maintaining breast identity in each breast cancer cell line category. As we have shown in other graphs, particularly for switches to the A compartment, more bins change to match lung in the metastatic vs. primary site cell lines. In most cases, more than 50% of the tissue-specific bins shift to look more like lung.

(5) Rest of the sections: The authors started to claim that the organ-specific metastasis permissive compartmental features mimic the destined end organ. The authors utilized additional non-breast cancer cell lines (prostate cancer cell lines LNCaP as localized and DU145 as brain metastatic) in brain metastasis to strengthen this claim. (DU145 in MetMap again is highly metastatic to lung, brain, and kidney). However, this makes one wonder that for cell lines that are capable of metastasizing to multiple organ sites (eg. MDA-MB-231, DU145, A459, H460), does it mean that they all acquire the permissive features for all these organs? This scenario is clinically relevant in Stage 4 patients who often present with not only one metastatic lesion in one single organ but multiple metastatic lesions in more than one organ (eg. concomitant liver and lung metastasis). Do the authors think that there might be different clones having different tropism-permissive 3D genome features or there might be evolutionary trajectory in this?

In my opinion, to further prove this point, the authors might need to consider doing in vivo experiments to collect paired primary and organ-specific metastatic samples to look at the 3D genome changes.

We agree that an ideal experimental follow up to this study would be to collect paired metastatic and primary tumors, either in mouse xenograft or, even better, from patients. This is beyond the scope of what we can do for our current paper, but we have added a statement to the discussion of further experiments that would be required to clarify this point.

(6) Technically, the study utilized public Hi-C data without generating new Hi-C data. The resolution of the Hi-C data for compartments was set at 250KB as the binning size indicating that the Hi-C data was at lower resolution so it might not be ideal to address other 3D genome architecture changes such as TADs or long-range loops. It is therefore unknown whether there might be permissive TAD/loop changes associated with organotropism and this is the limitation of this study.

Our decision to focus on A/B compartmentalization rather than TAD or loop structure in this analysis was intentional and biologically motivated, rather than solely being a reflection of data resolution. Both compartments and topologically associated domains (TADs) are key parts of genome organization and disruption of these structures has the potential to alter downstream gene regulation, as shown by numerous studies. However, compartments have been found, more so than TADs, to be strongly associated with cell type and cell fate. Therefore, in this manuscript, we decided to focus only on the compartment organization changes between different healthy and cancerous cells as they are more likely to represent the stable alterations of the genome organization malignant transformations.

(7) In the final sentence of the discussion the authors stated "Overall, our results suggest that genome spatial compartment changes can help encode a cell state that favors metastasis (EMT)". The "metastasis (EMT)" was in fact not clearly linked inside the manuscript. The authors did not provide a strong link between metastasis and EMT in their result description. It is also unclear whether the EMT-associated compartment identity would also correlate with the organotropic compartment identity.

We agree that this statement involves too strong of an assumption. The literature on this topic is vast and complex, and while there is abundant evidence that pathways of EMT can play important roles in facilitating metastasis, there are other pathways at play in the metastatic process as well (<https://journals.plos.org/Plosbiology/article?id=10.1371/journal.pbio.3002487>). We have made a clearer statement about this in the text now.

To address the question of whether the organotropic changes related to the EMT changes, we calculated the overlap between the genomic bins that strongly segregated cell lines in the

compartment principal component analysis (PC1) with those that showed “organotropic” changes. As you can see in supplementary table 3, this overlap is actually very small, where only 3% of bins are important both for the EMT segregation of cell lines and organotropism.

We have now included this overlap information as supplementary table 3 and have addressed this in the text.

Reviewer #2 (Public review):

Summary:

This work addresses an important question of chromosome architecture changes associated with organotopic metastatic traits, showing important trends in genome reorganization. The most important observation is that 3D genome changes consistent with adaptations for new microenvironments, including lung metastatic breast cells exhibiting signatures of the genome architecture typical to a lung cell-like conformation and brain metastatic prostate cancer cells showing compartment shifts toward a brain-like state.

Strengths:

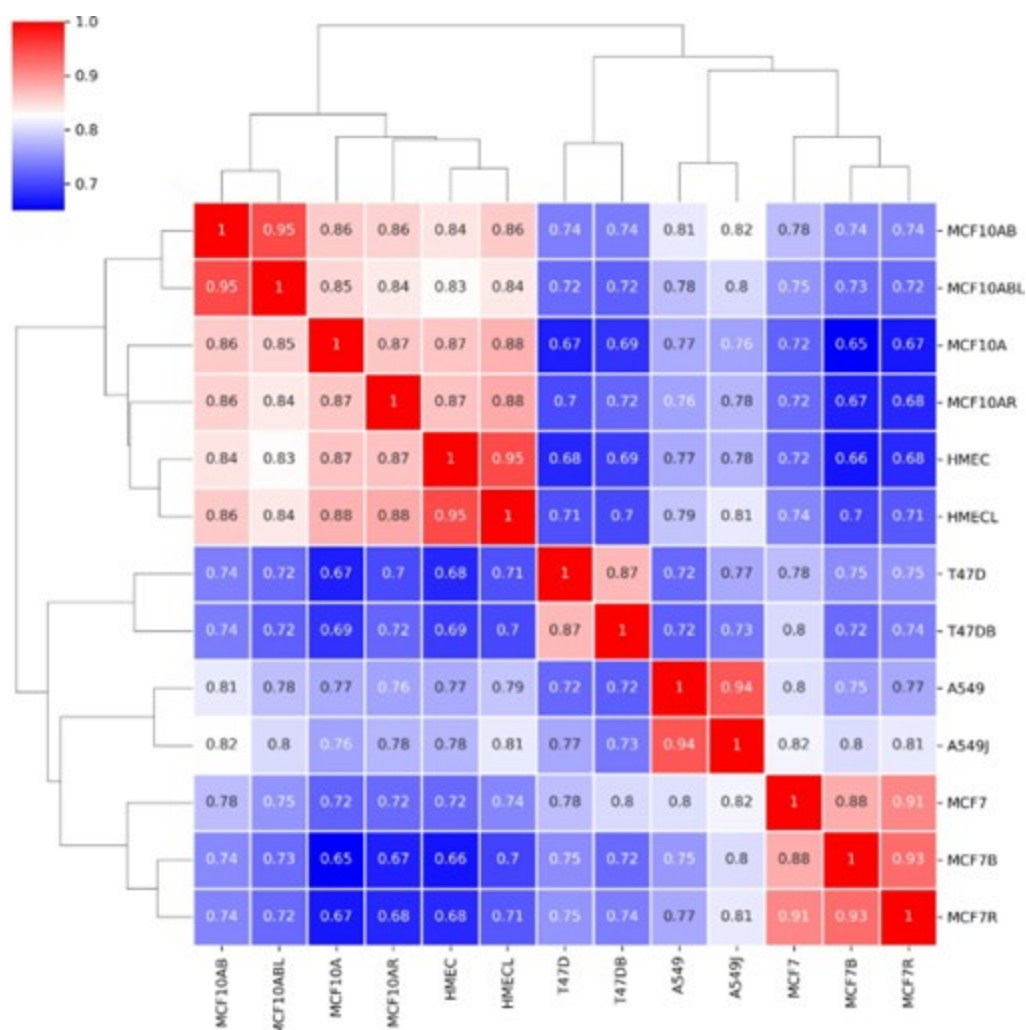
This work presents interesting original results, which will be important for future studies and biomedical implications of epigenetic regulation in norm and pathology.

Weaknesses:

The authors used publicly available data for 15 cell types. They should show how many different sources the data were obtained from and demonstrate that obtained results are consistent if the data from different sources were used.

In our revised version, we have provided a clarified table of information about all the publicly available data used from all the cell lines, indicating the sources of the data. The 17 datasets used come from 8 different studies. So, indeed, the reviewer is correct that many different sources of data were used. To address the question of whether our results would be consistent if data from different sources were used, we created a comparison map of the A/B compartment profiles for data from multiple sources when it was available. You can see below that the Hi-C data from different sources for the same cell lines cluster quite closely and show high correlation and are well separated from different cell lines. So, we do not think that source batch effects play a major role in our results.

Author response image 3.



Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Figure 1a: This figure could be re-formatted without the arrows. Arrows usually indicate upstream-to-downstream relationships along certain processes. Using arrows here would mislead people to think that the cell lines were derived from one another. The same could apply to the supplementary figures.

We have now edited figure 1a to include lines linking cell lines, indicating conceptual relationships, rather than arrows, which would imply direct derivation.

(2) Figure 1c: The PCA (PC2 axis) indeed seemed to separate the HER2 status quite well. One concern is MCF7, it is labeled as ERpos/HER2neg in MetMap but seems to be clustered as HER2pos in this study. Are they the same? (This again highlights the importance of cell line definition and annotation).

It is a good point that MCF7, while generally considered HER2 negative (we indicate this negative status in Supplementary Table 1), falls near HER2 positive cells in PCA space. This indicates that PCA captures tendencies but is not a perfect classifier. In a high dimensional,

complex system, it is expected that an unsupervised analysis such as this will not capture just one biological feature in a given principal component, and therefore something like HER2 status may not segregate perfectly. However, this analysis does suggest that MCF7 3D genome structure has features that are more similar to other HER2+ cell lines. This raises the interesting possibility that it may actually behave like HER2+ cells in some ways even while being HER2- itself. We have more clearly stated the MCF7 discrepancy in the text.

Reviewer #2 (Recommendations for the authors):

(1) The description of results can be shortened, to make it easier to read and understand.

In our revision, we have tried to clarify where possible, but it was difficult to shorten without losing important caveats and context (especially to make important points emphasized by reviewer 1).

(2) "100 most positive and negative eigenvalues for PC1" - please provide the correct description.

We have altered this to make it clearer and more correct: “using the genes from the regions with the top 100 most positive and 100 most negative eigenvector loadings for this PC1”

<https://doi.org/10.7554/eLife.103697.2.sa0>